Hadi Fares

Aerial Robotics & Autonomous Systems Researcher

hnf01@mail.aub.edu | linkedin.com/in/hadi-fares | Google Scholar

Education

American University of Beirut — Beirut, Lebanon
Bachelor's degree in Mechanical Engineering, GPA: 4.0/4.0 — *Aug. 2023 – May 2027*

Publications

- **Real-Time Elevation and Orientation-Aware Visual Localization for GNSS-Denied Drone Navigation.** *MDPI Drones*, Jun. 2026.
- **THC Vape Use Patterns and Device Features: Observations from User-Posted Videos.** Dec. 2025.

Experience

GNC Research Assistant, CraneAERO Team — Jan. 2026 – Present
Cranfield University — *United Kingdom*

- Member of CraneAERO, one of 11 global winners of the GoAERO Challenge, developing the CRANE (Cranfield Rapid Aerial Network for Emergency) autonomous eVTOL aircraft for search-and-rescue, disaster relief, and medical evacuation missions.
- Contribute to guidance, navigation, and control (GNC) design for autonomous VTOL flight, supporting safety, mission-readiness, and performance objectives.

Computer Vision & Embedded Systems Intern — Jun. 2026 – Present
Oreyeon

- Developing computer vision and embedded systems solutions, integrating perception pipelines with real-time hardware platforms.

Undergraduate Research Assistant, Aerial Robotics Lab — Jul. 2024 – Present
American University of Beirut — *Beirut, Lebanon*

- Develop autonomous UAV systems spanning perception, navigation, and flight control across multiple research projects in vision-based localization, multi-sensor fusion, and deep learning.
- Built an autonomous firefighting fixed-wing UAV with onboard fire detection and solution dispersal, and a high-speed racing drone currently in testing aiming for a world record.

Undergraduate Research Assistant, AI & Machine Learning Lab — Jul. 2024 – Present
American University of Beirut — *Beirut, Lebanon*

- Contribute to projects merging drones and aerial robotics with advanced AI, applying large language models (LLMs), neural networks, reinforcement learning (RL), and RLHF to autonomous systems.

Head of JEDIs (Junior Engineering Design Innovators) — Sep. 2024 – Present
AUB Makerspaces, Red Room — *Beirut, Lebanon*

- Lead engineering projects with students and professors, specializing in fabrication, electronics, and circuit design using advanced manufacturing tools.
- Manage Red Room operations, coordinate the JEDI team, and organize all makerspace events and workshops.
- Developed carbon-fiber-reinforced composites from recycled plastic bottles with optimized 3D-printing parameters.

Projects

Real-Time GPS-Denied Drone Navigation | *Python, PyTorch, FAISS, RLHF, OpenCV*

- Developed a visual localization system with autonomous window sizing and correlation-based orientation detection, achieving 97% accuracy and 59 FPS on preprocessed satellite maps.
- Implemented MobileNet-V3 embeddings with FAISS similarity search, validated across 75,000 frames and outperforming transformer methods; journal paper submitted for publication.

Autonomous Balloon-Popping Drone | *Python, OpenCV, YOLO, ROS, AirSim, ArduPilot, Raspberry Pi*

- Built an autonomous drone using onboard camera and ultrasonic sensors for real-time navigation, target detection, and approach control.
- Implemented YOLO-based red-balloon detection with lock-on tracking and path planning to actuate a robotic arm for interception; simulated in AirSim, deployed via ArduPilot and Raspberry Pi.

JetTracker AI | *Python, PyTorch, LSTM, OpenCV, Reinforcement Learning, Flask*

- Built a real-time fighter-jet tracking system using LSTM networks and reinforcement learning for interceptor path planning, deployed as a dockerized web application.

Technical Skills

AI & Machine Learning: Reinforcement Learning, RLHF, Multi-Agent RL, Graph Neural Networks, CNNs, LSTMs, Edge AI, PyTorch, TensorFlow
Computer Vision: Visual Localization, Image Stitching, Object Detection, YOLO, OpenCV
Autonomy & Robotics: SLAM, Sensor Fusion, Kalman Filters, Forward/Inverse Kinematics, Path Planning, Flight Control, GPS-Denied Navigation, ROS 2/Gazebo, PX4/ArduPilot
Embedded & Engineering: Embedded Systems, Python, C/C++, SolidWorks, ANSYS, Docker, 3D Printing